Lesson 35: Extensive-Form Games and Subgame Perfection

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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8 Extensive-Form Games

8.1 Game Trees

Definition 8.1 (Extensive-Form Game)**.** An *extensive-form game* is a tuple $\Gamma = (N, H, P, (\mathcal{I}_i)_{i \in N}, u)$ where:

(i) $N = \{1, \ldots, n\}$ is the set of players. We may include a "chance" player 0 (nature).

(ii) H is a set of *histories* (finite sequences of actions), including the empty sequence $\emptyset$ (the root). H is closed under prefixes: if $(a_1, \ldots, a_k) \in H$ then $(a_1, \ldots, a_l) \in H$ for all $l \leq k$. A history $h \in H$ is *terminal* if there is no a with $(h, a) \in H$. The set of terminal histories is $Z \subseteq H$.

(iii) $P : H \setminus Z \to N \cup \{0\}$ is the *player function*, assigning to each non-terminal history the player who moves at that point.

(iv) For each player $i \in N$, $\mathcal{I}_i$ is a partition of $\{h \in H \setminus Z : P(h) = i\}$ into *information sets*. If h, h' are in the same information set $I \in \mathcal{I}_i$, then the set of available actions $A(h) = A(h')$ is the same (denoted $A(I)$).

(v) $u = (u_1, \ldots, u_n) : Z \to \mathbb{R}^n$ assigns payoffs to terminal histories.

Definition 8.2 (Perfect Information)**.** An extensive-form game has *perfect information* if every information set is a singleton: $|I| = 1$ for all $I \in \mathcal{I}_i$, for all $i \in N$. In a game with perfect information, every player knows exactly which node the game has reached when it is their turn to move.

Definition 8.3 (Perfect Recall)**.** Player i has *perfect recall* if, for any two histories h, h' in the same information set $I \in \mathcal{I}_i$, player i's sequence of information sets and actions leading to h and h' is the same. Informally, no player forgets their own past actions or information.

Example 8.1 (Entry Game)**.** An incumbent (player 2) faces a potential entrant (player 1). The entrant first decides whether to *Enter* (E) or *Stay Out* (O). If the entrant enters, the incumbent decides whether to *Fight* (F) or *Accommodate* (A):

- O: payoffs $(0, 2)$.
- (E, F): payoffs $(-1, -1)$.
- (E, A): payoffs $(1, 1)$.

This is a game with perfect information (all information sets are singletons).

8.2 Strategies in Extensive-Form Games

Definition 8.4 (Pure Strategy in Extensive Form)**.** A *pure strategy* for player i in an extensive-form game is a function $s_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} A(I)$ satisfying $s_i(I) \in A(I)$ for each $I \in \mathcal{I}_i$.

Remark 8.1 (Strategies as Complete Contingent Plans). A strategy specifies an action at *every* information set, including those that cannot be reached given earlier choices. In the Entry Game (Example 8.1), player 2's strategy must specify what to do if the entrant enters, even if that node is never reached. This redundancy is essential for the normal-form representation and for defining subgame perfection.

Definition 8.5 (Behavioral Strategy)**.** A *behavioral strategy* for player i is a function $b_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} \Delta(A(I))$ satisfying $b_i(I) \in \Delta(A(I))$. That is, b_i assigns a probability distribution over actions at each information set, with independent randomization at different information sets.

Proposition 8.1 (Normal-Form Representation)**.** *Every extensive-form game has a corresponding normal-form game: the pure strategy sets are as in Definition 8.4, and the payoff $u_i(s_1, \ldots, s_n)$ is determined by tracing the unique path through the game tree when each player follows their strategy (resolving chance moves by expectation).*

Proof. Given a strategy profile $(s_1, \ldots, s_n)$, the unique path from the root to a terminal node is determined inductively: at each non-terminal history h, the next action is $s_{P(h)}(I)$ where I is the information set containing h (or a probability distribution if $P(h) = 0$, i.e., nature moves). Since the game tree is finite, this process terminates at some terminal history $z \in Z$. Define $u_i(s_1, \ldots, s_n)$ as the expected value of $u_i(z)$ where the expectation is over any chance moves. The resulting tuple $(N, (S_i), (u_i))$ is a well-defined normal-form game. □

Theorem 8.2 (Kuhn's Theorem)**.** *In a finite extensive-form game with perfect recall, every mixed strategy of player i has a* realization-equivalent *behavioral strategy: a behavioral strategy that induces the same probability distribution over terminal histories for every strategy profile of the other players. Conversely, every behavioral strategy has a realization-equivalent mixed strategy.*

Proof. We prove the direction from mixed to behavioral, which is the more useful one.

Let σ_i be a mixed strategy (a distribution over pure strategies) for player i in a game with perfect recall. We construct a behavioral strategy b_i as follows.

For each information set $I \in \mathcal{I}_i$ and action $a \in A(I)$, define

$$b_i(a|I) = \frac{\sigma_i\big(\{s_i : s_i(I) = a \text{ and } s_i \text{ reaches } I\}\big)}{\sigma_i\big(\{s_i : s_i \text{ reaches } I\}\big)},$$

where "s_i reaches I" means that the pure strategy s_i, together with some strategy profile of the opponents, leads to a history in I. In a game with perfect recall, whether s_i reaches I depends only on s_i's earlier actions, so this is well-defined. If the denominator is zero (no pure strategy in the support of σ_i reaches I), set $b_i(\cdot|I)$ arbitrarily.

The key property of perfect recall is that it makes the randomization at different information sets conditionally independent: knowing that player i reaches information set I does not reveal what σ_i prescribes at other information sets that are not on the path to I. This ensures that the behavioral strategy b_i produces the same realization probabilities (probability of reaching each terminal node) as σ_i.

The converse (behavioral to mixed) follows by defining $\sigma_i(s_i) = \prod_I b_i(s_i(I)|I)$, viewing each pure strategy as specifying an action at each information set. □

Behavioral Strategies and RL Policies

A behavioral strategy is exactly an RL policy: it maps information states (information sets) to probability distributions over actions. Kuhn's theorem justifies the standard RL assumption that we can restrict attention to behavioral strategies (policies) without loss of generality. The perfect recall condition is the Markov property in disguise: the agent's policy can depend on the current information state without needing to recall the randomization device used to reach that state.

9 Subgame Perfect Equilibrium

9.1 Motivation: Non-Credible Threats

Example 9.1 (Non-Credible Threat in the Entry Game)**.** In the Entry Game (Example 8.1), consider the strategy profile where player 1 plays O (stay out) and player 2 plays F (fight). The payoffs are $(0, 2)$. Is this a NE?

Player 1's payoff from deviating to E (enter): given player 2's strategy F, the payoff is $-1 < 0$. So player 1 has no incentive to deviate.

Player 2's strategy F is not tested because player 1 stays out, so player 2's information set is not reached. Player 2 trivially has no incentive to deviate (any strategy gives the same outcome of $(0, 2)$ when player 1 plays O).

Therefore (O, F) is a NE. But it relies on a *non-credible threat*: player 2 threatens to fight, but if the entrant actually entered, fighting gives player 2 a payoff of -1 versus 1 for accommodating. Rational player 2 would never actually fight. The Nash equilibrium concept fails to exclude such non-credible threats because it does not require rational behavior at unreached information sets.

9.2 Subgames and Subgame Perfection

Definition 9.1 (Subgame)**.** A *subgame* of an extensive-form game Γ beginning at history $h \in H \setminus Z$ is the restriction of Γ to the histories that extend h, provided:

(i) h is a singleton information set: $\{h\}$ is an information set for player $P(h)$.

(ii) Every information set that contains a successor of h is entirely contained within the subtree rooted at h.

The subgame includes all histories $h' \in H$ such that h is a prefix of h', with the same player function, information sets (restricted to the subtree), and payoffs.

Remark 9.1. In a game with perfect information, every non-terminal history h begins a subgame (since all information sets are singletons). In games with imperfect information, subgames may be rare — the entire game is always a subgame (beginning at the root).

Definition 9.2 (Subgame Perfect Equilibrium)**.** A strategy profile s^* is a *subgame perfect equilibrium* (SPE) if its restriction to every subgame is a Nash equilibrium of that subgame.

Proposition 9.1 (SPE Refines NE)**.** *Every subgame perfect equilibrium is a Nash equilibrium.*

Proof. The entire game is itself a subgame (beginning at the root). If s^* is an SPE, then it is a NE of the entire game by definition. □

Proposition 9.2 (SPE in the Entry Game)**.** *The Entry Game (Example 8.1) has a unique SPE: (E, A) (Enter, Accommodate) with payoffs $(1, 1)$.*

Proof. The game has two subgames: the entire game (rooted at the empty history) and the subgame starting after player 1 chooses E (where player 2 decides between F and A).

In the subgame after E: player 2 chooses between F (payoff -1) and A (payoff 1). The unique NE of this subgame is A.

Given that player 2 plays A in the subgame, player 1 in the full game chooses between O (payoff 0) and E (payoff 1, since player 2 accommodates). Player 1 chooses E.

The unique SPE is (E, A). The non-credible equilibrium (O, F) is eliminated because F is not a NE in the subgame after entry. □

9.3 Backward Induction

Definition 9.3 (Backward Induction)**.** For a finite game with perfect information, *backward induction* is the following procedure:

1. Identify all "penultimate" nodes: non-terminal histories h such that all successors (h, a) are terminal. At each such h, the player $P(h)$ chooses the action maximizing their payoff: $a^*(h) = \operatorname{argmax}_{a \in A(h)} u_{P(h)}(h, a)$.
2. Replace each penultimate node h with a terminal node having payoff $u(h, a^*(h))$.
3. Repeat until the root is reached.

The resulting strategy profile specifies an action at every node of the original tree.

Theorem 9.3 (Backward Induction Yields SPE)**.** *In a finite game with perfect information and no ties (i.e., the payoff function is injective when restricted to any player's choices at any node), backward induction produces the unique subgame perfect equilibrium.*

Proof. We prove by strong induction on the depth d of the game tree (the maximum number of actions in any history).

Base case ($d = 1$): The game consists of a single player choosing an action at the root, with all successors being terminal. The backward induction strategy (choose the payoff-maximizing action) is trivially the unique NE and SPE.

Inductive step: Assume the result holds for all games of depth $< d$. Consider a game Γ of depth d with perfect information and no ties.

At the root, player $P(\emptyset)$ chooses action $a \in A(\emptyset)$. Each action a leads to a subgame Γ_a of depth $< d$. By the inductive hypothesis, each Γ_a has a unique SPE s_a^* with a well-defined payoff vector $v(a)$.

In the full game, backward induction assigns to the root the action $a^* = \operatorname{argmax}_{a \in A(\emptyset)} v_{P(\emptyset)}(a)$, which is unique (no ties).

We show the resulting strategy profile s^* is the unique SPE.

s^* is an SPE: In any subgame Γ_a, the restriction of s^* is s_a^*, which is an SPE of Γ_a by the inductive hypothesis. It remains to check that no player wants to deviate at the root. But the root player $P(\emptyset)$ anticipates the SPE outcome in each subgame (by the inductive hypothesis, the SPE outcome is unique), and a^* maximizes over these outcomes. The other players do not move at the root.

Uniqueness: Any SPE $\hat{s}$ must restrict to the unique SPE in each subgame Γ_a (by the inductive hypothesis), so the root player faces the same payoffs $v(a)$ and, with no ties, must choose a^*. Hence $\hat{s} = s^*$. □

9.4 Zermelo's Theorem

Zermelo's Theorem

Theorem 9.4 (Zermelo, 1913)**.** *Every finite extensive-form game with perfect information has a pure strategy subgame perfect equilibrium. If the game also has no ties (each player has distinct payoffs at all terminal nodes reachable from any decision node), the SPE is unique.*

Proof. **Existence.** We prove by strong induction on the number of non-terminal histories $|H \setminus Z|$.

Base case: If $|H \setminus Z| = 1$, the game has a single decision node (the root) followed by terminal nodes. The player at the root has at least one action maximizing their payoff (the strategy set is finite and non-empty). Any such action constitutes a pure strategy SPE.

Inductive step: Let Γ have $|H \setminus Z| = k > 1$. The root player $P(\emptyset)$ has actions $A(\emptyset) = \{a_1, \ldots, a_l\}$, and each action a_j leads to a subgame Γ_{a_j} with strictly fewer non-terminal histories. By the inductive hypothesis, each Γ_{a_j} has a pure strategy SPE $s_{a_j}^*$ with payoff vector $v(a_j) \in \mathbb{R}^n$.

Define the strategy profile s^* as follows: at the root, choose $a^* \in \operatorname{argmax}_{a_j} v_{P(\emptyset)}(a_j)$ (this set is non-empty since l is finite). In each subgame Γ_{a_j}, follow $s_{a_j}^*$.

We verify s^* is an SPE. For any subgame not containing the root, the restriction of s^* is $s_{a_j}^*$ for the appropriate j, which is an SPE by hypothesis. For the entire game (which is also a subgame): the root player is choosing optimally given the SPE continuations (by construction), and no other player moves at the root, so no player has a profitable deviation. Hence s^* is an SPE.

Uniqueness under no ties. This follows from the backward induction argument in Theorem 9.3: at each node, the argmax is unique when payoffs are distinct, so the SPE is unique. □

Corollary 9.5 (Chess Has a Determined Outcome)**.** *In chess (viewed as a finite game with perfect information — the 50-move rule and threefold repetition ensure finiteness), exactly one of the following holds: White has a winning strategy, Black has a winning strategy, or both players can force a draw.*

Proof. Chess is a two-player, finite, perfect-information game with three outcomes: White wins, Black wins, or draw. By Zermelo's theorem, a pure strategy SPE exists. In this SPE, the outcome is determined: it is either a White win, a Black win, or a draw. The SPE strategy for the favored player (or both in the draw case) is the "winning" (or "drawing") strategy that guarantees this outcome regardless of the opponent's play. □

Backward Induction and Dynamic Programming

Backward induction in game theory is the multi-agent analog of dynamic programming in single-agent MDPs. In both cases, we work backward from terminal states, computing optimal decisions at each stage given the optimal continuation. The difference is:

- **Single-agent DP**: The Bellman equation $V(s) = \max_a[r(s,a) + \gamma V(s')]$ has a unique optimal value function (contraction mapping theorem).
- **Multi-agent backward induction**: At each node, the "optimal" action depends on the other player's SPE strategy, making it a game-theoretic rather than optimization problem. With perfect information and no ties, the SPE is still unique. With imperfect information or ties, multiplicity arises.

AlphaGo and AlphaZero use MCTS (Monte Carlo Tree Search) to approximately perform backward induction in game trees too large for exact computation.

9.5 Behavioral Strategies vs. Mixed Strategies in Extensive Form

Remark 9.2 (Kuhn's Theorem Revisited). Theorem 8.2 establishes that in games with perfect recall, we can work with behavioral strategies (RL policies) without loss of generality. This is crucial for computational purposes: a behavioral strategy specifies a distribution at each information set independently, requiring $O(\sum_I |A(I)|)$ parameters, whereas a mixed strategy over pure strategies may require exponentially many parameters (one probability for each complete contingent plan).

Proposition 9.6 (Failure Without Perfect Recall)**.** *In games without perfect recall, mixed and behavioral strategies may yield different sets of reachable outcome distributions. There exist games without perfect recall in which a mixed strategy NE exists but no behavioral strategy NE does.*

Proof. Consider a one-player game where the player moves twice but forgets their first action at the second move. Specifically: the player first chooses L or R, then (in a single information set containing both post-L and post-R histories) chooses A or B. Payoffs:

- (L, A): payoff 1, (L, B): payoff 0,
- (R, A): payoff 0, (R, B): payoff 1.

A behavioral strategy specifies $p = \mathbb{P}(\text{choose } L)$ at the first node and $q = \mathbb{P}(\text{choose } A)$ at the second information set (which contains both post-L and post-R). The expected payoff is $pq + (1-p)(1-q) = 1 - p - q + 2pq$. The maximum over behavioral strategies is $1/2$ (achieved, e.g., at $p = q = 1/2$).

However, the mixed strategy that plays (L, A) with probability $1/2$ and (R, B) with probability $1/2$ achieves expected payoff $1/2 \cdot 1 + 1/2 \cdot 1 = 1 > 1/2$. This mixed strategy has no realization-equivalent behavioral strategy, since the behavioral strategy cannot correlate the first and second moves. □

10 Exercises

Exercise 10.1 (Pure NE Enumeration)**.** Find all pure strategy Nash equilibria of the following game:

	L	C	R
T	$(2,1)$	$(0,4)$	$(1,2)$
M	$(3,2)$	$(1,1)$	$(0,3)$
B	$(1,0)$	$(2,3)$	$(3,1)$

For each NE, verify the best response condition.

Exercise 10.2 (IESDS Application)**.** Apply iterated elimination of strictly dominated strategies to the game:

	L	C	R
T	$(3,1)$	$(0,0)$	$(1,3)$
M	$(1,2)$	$(2,4)$	$(3,0)$
B	$(0,3)$	$(4,2)$	$(2,1)$

At each step, identify the dominated strategy and the strategy that dominates it. State the surviving strategy profiles.

Exercise 10.3 (Mixed NE in 2×2 Games)**.** Compute all Nash equilibria (pure and mixed) of the Stag Hunt (Example **??**). Verify the indifference principle for the mixed NE.

Exercise 10.4 (Minimax Computation)**.** For the zero-sum game with payoff matrix

$$A = \begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix},$$

compute the value of the game and the optimal mixed strategies for both players. Formulate the problem as a linear program and verify the minimax theorem directly.

Exercise 10.5 (Correlated Equilibrium Construction)**.** Consider the game of Chicken:

	D	S
D	$(-5,-5)$	$(2,-1)$
S	$(-1,2)$	$(0,0)$

(a) Find all pure and mixed Nash equilibria.
(b) Construct a correlated equilibrium that Pareto-dominates the mixed NE.
(c) Verify that your CE satisfies the incentive-compatibility constraints.

Exercise 10.6 (Backward Induction)**.** Consider the following three-stage perfect-information game: Player 1 chooses A or B. If A, player 2 chooses C or D. If B, player 2 chooses E or F. If D, player 1 chooses G or H. Terminal payoffs (player 1, player 2):

- $A \to C$: $(3,2)$, $A \to D \to G$: $(4,1)$, $A \to D \to H$: $(1,3)$,
- $B \to E$: $(2,4)$, $B \to F$: $(5,0)$.

Apply backward induction to find the SPE. List the complete strategy for each player.

Exercise 10.7 (Non-Credible Threats)**.** Consider a modified entry game: an entrant chooses E or O. If E, the incumbent chooses F, A, or P (where P means "price war"). Payoffs:

- O: $(0,3)$, (E,F): $(-2,-1)$, (E,A): $(1,1)$, (E,P): $(-3,-2)$.

(a) Find all pure strategy NE of the normal-form representation.
(b) Identify which NE involve non-credible threats.
(c) Find the unique SPE.

Exercise 10.8 (Kuhn's Theorem Application)**.** In a two-player signaling game, player 1 (the sender) observes a private type $t \in \{H, L\}$ (each equally likely) and sends a message $m \in \{h, l\}$. Player 2 (the receiver) observes the message but not the type, and chooses an action $a \in \{U, D\}$. Player 1 has four pure strategies (mapping types to messages).

(a) How many pure strategies does player 1 have? Player 2?
(b) Write down the normal-form representation.
(c) Verify that this game has perfect recall.
(d) Construct a behavioral strategy for player 1 and its realization-equivalent mixed strategy.

Exercise 10.9 (Properties of NE)**.** Prove or disprove the following statements for finite games:

(a) Every game with a dominant strategy equilibrium has a unique NE.
(b) If IESDS yields a unique strategy profile, it is a NE.
(c) The set of NE is always convex.
(d) In a two-player zero-sum game, if (x^*, y^*) and (x^{**}, y^{**}) are both NE, then (x^*, y^{**}) is also a NE.

Exercise 10.10 (SPE Existence Without Perfect Information)**.** Construct a two-player extensive-form game with imperfect information that has:

(a) Exactly one subgame (the entire game).
(b) A NE that is not "sequentially rational" at the non-singleton information set.

This illustrates why SPE has limited refinement power in games with imperfect information, motivating stronger concepts like sequential equilibrium.

11 Summary of Key Results

Result	**Statement**	**Ref.**
Order Independence of IESDS	In finite games, the set of strategies surviving iterated elimination of strictly dominated strategies is independent of the elimination order.	Thm. ??
NE as Mutual Best Response	s^* is a NE iff $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all i.	Prop. ??
Indifference Principle	In a mixed NE, all pure strategies in the support yield equal expected payoff; strategies outside the support yield weakly less.	Thm. ??
Nash's Existence Theorem	Every finite game has at least one mixed strategy NE. Proved via Kakutani's fixed point theorem.	Thm. ??
Kakutani's Fixed Point Thm	A uhc correspondence with non-empty convex values on a compact convex set has a fixed point. Proved via Brouwer.	Thm. ??
Von Neumann's Minimax Thm	$\max_x \min_y x^\top Ay = \min_y \max_x x^\top Ay$ for any matrix A. Proved via LP duality.	Thm. ??
NE in Zero-Sum Games	NE strategies are interchangeable and all yield the game value.	Cor. ??
NE $\subseteq$ CE	Every NE (as a product distribution) is a correlated equilibrium.	Thm. ??
CE via LP	The CE set is a convex polytope; optimizing over it is an LP.	Prop. ??
Kuhn's Theorem	With perfect recall, mixed and behavioral strategies are realization-equivalent.	Thm. 8.2
SPE Refines NE	Every subgame perfect equilibrium is a Nash equilibrium.	Prop. 9.1
Backward Induction $\Rightarrow$ SPE	In finite perfect-information games with no ties, backward induction yields the unique SPE.	Thm. 9.3
Zermelo's Theorem	Every finite perfect-information game has a pure strategy SPE. Unique if no ties.	Thm. 9.4

Looking ahead. This lesson establishes the static and single-shot extensive-form game theory needed for multi-agent RL. The next game theory topics — repeated games, stochastic games, and mechanism design — build directly on these foundations. In Phase 02 (Core RL), the MDP framework can be viewed as a single-agent "game against nature." In Phase 05 (MARL), stochastic games combine MDPs with the multi-player game-theoretic structure developed here: at each state, agents play a stage game whose solution concept (Nash, minimax, correlated) determines the Bellman operator for the multi-agent value function.